

Predictive Analysis for Risk of Stroke Using Machine Learning Techniques

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Abstract— Stroke occurs when our brain's blood flow is stopped or reduced, restricting brain tissue from receiving oxygen and important nutrients. Treatment of stroke disease is very crucial. So, the prediction of stroke is significant for early intervention and treatment. Stroke can be predicted by analyzing different warning signs. In this experiment, we implement a process of stroke risk prediction from our dataset using the various machine learning algorithms. Data imputation, feature selection, data preprocessing is considered as the initial job in this work. Some physical parameters such as heart disease, age, BMI, gender, hypertension, etc. are considered as the feature is used to model training and testing. In this study, adaBoost classifier, artificial neural network, decision tree classifier, k-nearest neighbour (KNN) classifier, random forest, stochastic gradient descent (SDG), support vector machine (SVM), XGBoost classifier are implemented to predict the risk possibility of stroke. Then voting classifier is implemented on these eight traditional classifiers. After analyzing different machine learning algorithms with voting classifiers, we found 98% accuracy to predict the risk factor of stroke, which is better than other models.

Keywords— Stroke, Machine learning, Classification, Data preprocessing, Confusion matrix

I. INTRODUCTION

The world is on the start of the fifth industrial revolution. A great revolution in modern technology has occurred in the health sector. However, some diseases cannot be completely cured, even with modern medical equipment or medicine. Stroke is the most heinous diseases in the modern world. A clot or a burst blocks a brain blood vessel, oxygen, and other nutrients cannot enter the brain [1]. As a result, brain cells die, and a stroke occurs. It attacks the arteries or vein that led to and from the brain. Stroke is now a silent killer disease in Bangladesh, 21.5% of all aged people in Bangladesh are affected by stroke among all diseases [2]. In 2016, strokes are the third biggest reason of death and 5.7 million deaths, 13% of all worldwide deaths [3]. After 2016, however, WHO deemed stroke the second most important reason of death and also the third most significant reason of disability. In low and medium-income nations around the world, 70% of stroke and 87% of stroke-related death occur [4]. World Health Organization (WHO) reported that, in every 4-5 minutes affected persons are departed [5].

Stroke can be classified in two ways; one is ischemic stroke another one is hemorrhagic stroke. In an ischemic stroke, the blood vessel is obstructed by the pressure of clot, and on the other side a hemorrhagic stroke, comparatively weak blood artery explodes and bleeds around the brain. Stroke disease can be minimized by a healthy and balanced lifestyle that is wiping off the unhealthy lifestyle such as smoking, alcoholic drink, controlling body weight and glucose consumption and keep up good health of body organ. Possibility of stroke disease is very important that's why early treatment can reduce permanent injury of organ or mortality. Machine learning (ML) play an important part within the

deciding processes in stroke prediction. When a stroke occurs, identify the affected patient's condition and begin therapy within a half hour or one hour is very important. Otherwise, saving the patient's life becomes tough. It can be difficult to predict stroke based on risk factors since risk variables are complicated and inconsistent [6]. In this experiment, we compare some machine learning methods which may be predicted stroke early based on physical characteristics. Previously, the most common are used to predict stroke. However, in this paper, we utilized various classifiers such as AdaBoost Classifier, Artificial Neural Network, KNN Classifier, SGD, SVM, XGBoost, etc. The predicted output of our models shows that the suggested approach can apply to bio-signals in real-time.

II. LITERATURE REVIEW

Several related works have been done to predict stroke previously. Some of the associated results are:

Aditya Khosla et al. [7] presented a feature extraction method which find strong features using their proposed method: conservative mean. They compared their work Cox Proportional Hazards model with different machine learning process to find stroke possibility using well known Cardiovascular-Health-Study (CHS) dataset.

Jaehak, et al. [8] discussed an AI model for prediction of stroke illness based on the real-time diagnostic procedure. They proposed a stroke prediction process which can identify stroke victimization period biological-signals using artificial intelligence system.

Singh, M. Sheetal, and Prakash Choudhary [9] proposed a stroke prediction method using artificial intelligence. They compared different models for stroke prediction using the Cardiovascular-Health-Study (CHS) dataset. Authors used a decision tree model for selecting important features, a principal part analysis algorithmic program for minimizing the complexity and endorse a backpropagation artificial neural network algorithmic program, for build a classifying model.

Pradeepa S. et al. [10] proposed a method to detect stroke disease risk factors using a ML technique. Their vision was determining the different symptoms related to the stroke and proposed a preventive method for stroke using a DRFS methodology. Health parameters are vital for detecting risk factors of stroke.

Chiun-Li, et al. [11] developed an automated stroke prediction model based on CNN method. Their design removed the unaffected space that's not suffering from the stroke in the CT scan image of the affected brain. They selected modified images and used the Data Augmentation method. After that, they train and test the model using CNN. Cheon et al. [12] proposed a neural network supported method for predict stroke. They used the principal-component-analysis (PCA) method to feature quantile scaling and compared their method to five other ML methods.

MU Emon et al. [13] introduced ten ML classification methods to predict stroke. They achieved high accuracy in their proposed weighted voting classifier.

III. RESEARCH METHODOLOGY

The method of this proposed work is described in this part. The methodology part is divided into collection of data, description of data, pre-processing of dataset and implementation of the different machine learning classifiers. Fig. 1 representing the working process implemented in our work.

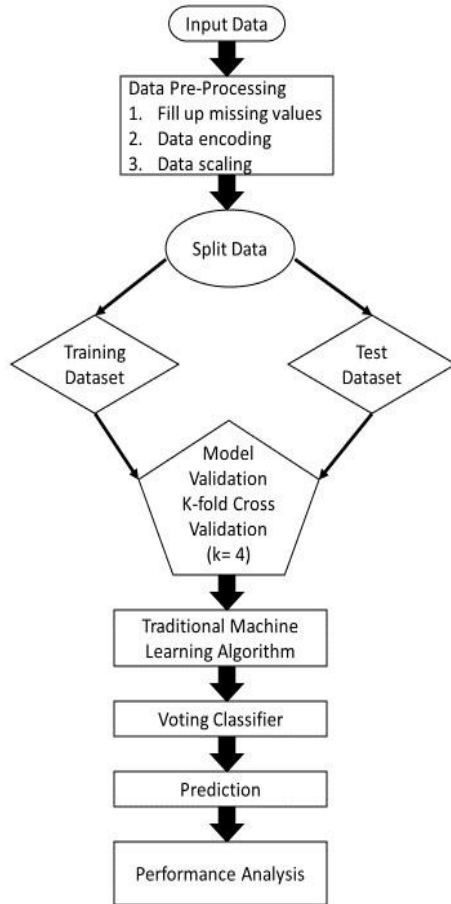


Fig. 1. Schematic working process diagram of our system

The systematic working procedure are mentioned in below:

A. Collection of Data and Description of Data:

Stroke disease dataset is collected from the different hospitals in Bangladesh. This dataset has 8600 subjects where 2500 subjects were experience stroke. Each data includes 11 attributes. Here some attribute has numerical data, and some attribute has categorical data. Feature age is our subject age and which is numerical value. Feature hypertension is that this subject has hypertension or no hypertension and this is containing numerical value. Feature average glucose is the level of subject glucose consumption and that contain numerical value. Feature BMI is ratio of subject height & subject weight and that contain numerical value. Feature smoke means a subject's smoking condition and that contain categorical value. Feature stroke indicate the subject had a stroke or not had stroke in the past and that contain numerical value. Feature gender is the subject's gender and that is categorical data. Feature work type is the subject's employment status and that contain categorical value. Feature

residence_type is the subject residence condition and that contain categorical value. Feature ever married represent subject's marital status and that contain categorical value. Feature stroke is the decision class and the remaining attributes are the input class.

B. Data Pre-processing:

To make the dataset effective, at first, we look for the null value and duplicate value. Duplicate data were eliminated from the dataset. Then null values in the dataset were fill up by mean values and median values. The missing numerical data were fill up using mean value of that attribute, and the categorical null data were fill up by median value of that attribute. Using One Hot Encoding method, we convert all categorical values into numeric form. When dealing with categorical data for ML algorithms, encoding is mandatory in preprocessing stage. We use one-hot encoding because the ML algorithm cannot handle categorical data. Then we normalize all the numeric values. After that, we divide our dataset in two groups, one group is the training dataset and other group is the testing dataset. Training data and testing data was employed to train and test the model respectively to observe machine learning model performance.

C. Machine Learning Classifier:

In our paper, we have used nine different machine learning classifiers. The classifiers are AdaBoost, ANN, Decision Tree, KNN, Random Forest, SGD, SVM, XGBoost. All these classifiers are well known, and we can compare our results with other similar research work. We use 75% data of our dataset to train our algorithm, and the remaining 25% of the data is considered to assess the trained model. For ML model validation, we use k-fold cross validation process. In the k-fold cross validation technique, total dataset is employed to training and testing the classification process. The dataset is splitted into k parts, that known as fold. In training procedure, this process uses k-1 folds to train ML model and one-fold is employed to test model. This process is repeated k times, and every fold can be considered as test data-set. The ultimate outcome is the common for the testing group performance using all the used folds. Advantage of this technique, that the all samples within the data-set are used for train and test, which removes the high variance. Confusion matrices are used to evaluate the model's performance by calculating accuracy, recall, precision, f-1 score, false-positive rate, false-negative rate. Analyzing these values, we find out the best model to predict stroke.

IV. RESULT ANALYSIS

A. Correlation Matrix:

A correlation matrix is used to determine the relationship between stroke attributes and other attributes. Fig. 2 represents the correlation matrix among features of the dataset. From the figure, among the features of the dataset age, hypertension, gender, avg glucose level, BMI, smoking status, heart disease has a significant consequence on stroke, and ever married, work_type, residence_type has a more negligible effect on the stroke. The previous study of medical science also represents that hypertension, heart disease, glucose level, age, BMI, smoking has a great impact on stroke [14]. That's why negligible features were removed.

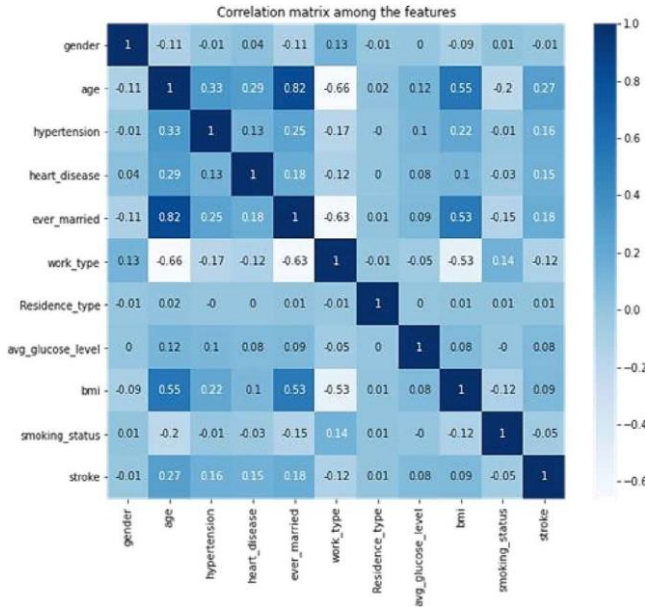


Fig 2. Confusion matrix among the dataset attribute

B. Model Evaluation:

We've used multiple algorithms and gained much better accuracy. 2150 data is used for the testing purpose of these models among 8600 data. For preprocessing and the development of the classification model, we utilize python ecosystems, jupyter notebook, sci-kit-learn, pandas, NumPy, matplotlib library.

TABLE I. CONFUSION MATRIX OF THE DIFFERENT MACHINE LEARNING MODELS

Model	Predicted →	No Stroke	Stroke
	Actual ↓		
AdaBoost	No Stroke	1286	324
	Stroke	139	401
ANN	No Stroke	1394	235
	Stroke	143	378
Decision Tree	No Stroke	1506	65
	Stroke	77	502
KNN	No Stroke	1370	131
	Stroke	171	478
Random Forest	No Stroke	1557	101
	Stroke	88	404
SGD	No Stroke	1331	195
	Stroke	128	496
SVM	No Stroke	1321	154
	Stroke	113	562
XGBoost	No Stroke	1260	157
	Stroke	213	520
Voting Classifier	No Stroke	1532	32
	Stroke	17	519

The confusion matrix of the various classifiers is represented in Table I. The confusion matrix of a machine learning model analyzes the target with predicted value. From confusion matrix, we find true-negative (TN), false-positive (FP), false-negative (FN) and true-positive (TP) values. Using TN, FP, FN, TP value we calculated accuracy, precision, recall, f-1 [15] scores as follows:

Accuracy means correctly classified cases divided by the total of test cases and given by:

$$Accuracy = \frac{TP+TN}{TP+TN+FN+FP} \quad (1)$$

The fraction of accurately identified instances in actually positive cases is referred to as precision and given by:

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

The recall means all correctly classified cases. In the disease prediction, recall value is critical and is given by:

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

The f-1 score is an average of estimated precision and revocation that takes both accuracy into account. The f-1 score is expressed as:

$$f-1 \text{ score} = 2 * \frac{Recall * Precision}{Recall + Precision} \quad (4)$$

C. Performance Analysis:

To prevent the bias of the dataset k-fold cross-validation was used. The ten-fold cross-validation approach was employed in this research. Besides, for each model, we estimated the false positive and false negative rates. FP and FN rate is a concern for any model. A false-positive case may lead to false treatment. A false-negative case may lead to a false diagnosis, which is a very serious problem if the disease is ignored. We found a very low rate of FP and FN in the voting classifier, which makes our model more acceptable. Table II measures the accuracy, precision, recall and f-1 score of different classification methods where the formula is given in equations (1), (2), (3), and (4). We utilized the voting classifier to evaluate the model's performance after employing eight typical machine learning classifiers.

A voting classifier is a machine learning model that trains from an aggregation of many models and estimates an output based on the possibility of the output class. We got an accuracy of 98% in the voting classifier, 93% in the decision tree, 91% in the random forest, 88% in support vector machine, 86% in KNN, 85% in stochastic gradient descent, 83% in XGBoost, 82% in ANN and 78% in AdaBoost. Because it calculated the outcome based on the certainty of output in these eight classifiers, the voting classifier had the highest accuracy rate among models.

Precision is known as proportion of accurately predicted positive observations to all expected positive observations. A high accuracy rate corresponds to a low false-positive rate. We got the highest precision rate in the voting classifier that is 0.942 and the lowest precision in AdaBoost is 0.788.

The recall is the proportion of accurately predicted positive case to the all cases in the class. In our research, all of the models had a high recall value.

The F-1 score is calculated as the average of precision and recall. As a result, this score considers both false-positives and false-negatives. Although it is not as apparent as accuracy, f-1 is frequently more useful than accuracy, especially if the class distribution is unequal. We got the highest f-1 score in the voting classifier that is 0.955 and the lowest in AdaBoost that is 0.634. Fig. 3 represents the visualization of the stroke risk prediction accuracy, recall, precision, and f-1 score for all types of machine learning classifiers.

Table III represents some related works to predict stroke using different machine learning approaches, which have been done previously. We evaluated certain state of the art approaches and their accuracy with our suggested model, and the proposed research attained a 98% accuracy.

TABLE II. PERFORMANCE ANALYSIS OF DIFFERENT CLASSIFICATION METHOD

Model	accuracy	precision	recall	f-1 score	FP rate	FN rate
AdaBoost	78%	0.533	0.743	0.634	20%	26%
ANN	82%	0.617	0.726	0.667	14%	27%
Decision Tree	93%	0.885	0.867	0.876	4%	13%
KNN	86%	0.785	0.737	0.760	9%	26%
Random Forest	91%	0.800	0.821	0.810	6%	18%
Stochastic Gradient Descent	85%	0.718	0.795	0.754	13%	21%
Support Vector Machine	88%	0.785	0.833	0.808	10%	17%
XGBoost	83%	0.738	0.709	0.738	11%	29%
Voting Classifier	98%	0.942	0.968	0.955	2%	3%

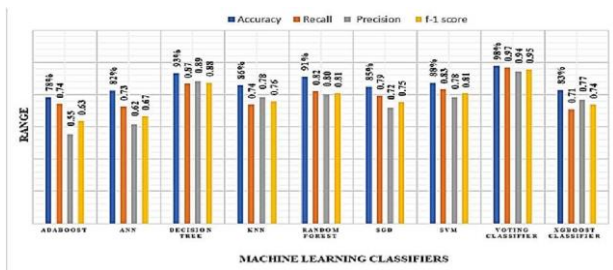


Fig 3. The accuracy, recall, precision, and f-1 score of different classification method

TABLE III. RESULT COMPARISON OF THE RELATED WORK

Reference	Algorithm	Accuracy
Singh et al. [9]	ANN	96%
Chiun-Li et al. [11]	CNN	90%
Cheon et al. [12]	DNN	83%
Amini et al. [16]	KNN	94%
Cheng et al. [17]	ANN	95%
Govindarajan et al. [18]	NPL-ML	95%
This work	Voting classifier	98%

The objective of this analytic research is to predict stroke risk and assess which method delivers the highest accuracy. Finally, the get voting classifier is the best algorithm to predict the stroke at an early stage.

V. CONCLUSION

This paper predicts the risk of strokes based on the classification approach. Eight different learning machines have been used to this data set, which reveals a 98% accuracy rate in voting classifier than other well-known approaches for prediction of stroke risks. As a consequence, voting classifier might be used to determine the risk of stroke. If a stroke can be predicted at an early stage, it is beneficial to the patient to prevent the stroke. Thus, this approach can be used in healthcare to identify the risk of stroke. In the future, brain imaging such as computed tomography and MRI may be combined as a feature to improve model performance. IoT system can be added in this system where the sensor will send the reading data to the cloud server. The machine learning model is also incorporated so that the webserver automatically receives the predicted accuracy.

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